

APM & *Advanced KPIs*

An intermediate-to-advanced path through distributed tracing, business observability, and SLO engineering with Datadog — engineered into four focused days.

4

DAYS

12

MODULES

30+

HANDS-ON LABS

40+

KPIS

FACILITATED BY **STACKMASTER TECH**

INTERMEDIATE → ADVANCED

From traces to business outcomes

This condensed program moves beyond infrastructure metrics into the engineering of application performance and business value. Over four days, participants master distributed tracing, advanced trace and latency analytics, business-KPI instrumentation, digital-experience monitoring, SLO engineering, and AI-driven observability — closing with a full-stack capstone that ties every signal to outcomes.

INTENDED AUDIENCE

SRE Engineers

DevOps Engineers

Platform Engineers

Application Support Teams

Observability Engineers

Architects

FOCUS AREAS

Application Performance Monitoring

Distributed Tracing

Business KPIs

SLO Engineering

User Experience Monitoring

Advanced Analytics

AI-Driven Observability

4

DAYS

12

MODULES

30+

LABS

40+

KPIS

6

KPI DOMAINS

THE FOUR-DAY ARC

01

Advanced APM & Distributed Tracing
Observability foundations, APM architecture, service maps.

02

Application Performance Engineering
Trace analytics, performance KPIs, continuous profiling.

03

Business KPIs & Digital Experience
Business observability, RUM, synthetic monitoring.

04

Reliability, AIOps & Executive Reporting
SLO engineering, intelligent alerting, AIOps, executive dashboards, full-stack capstone.

DAY 01

Advanced APM Foundations & Distributed Tracing

Ground the practice — application observability, Datadog APM architecture, and end-to-end service dependency analysis.

3 Modules · 2 Hands-On Sessions

01 • Modern Application Observability

MONITORING EVOLUTION

Infrastructure Monitoring

Application Monitoring

Full-Stack Observability

Business Observability

THREE PILLARS

Metrics

Logs

Traces

WHY APM MATTERS

User Experience

Revenue Impact

MTTR Reduction

Service Reliability

BUSINESS IMPACT OF PERFORMANCE

Revenue Loss due to Latency

Impact of Slow APIs

Impact of Failed Transactions

02 • Datadog APM Architecture

APM COMPONENTS

Trace Agent

Application Instrumentation

Service Mapping

Distributed Tracing

TRACE LIFECYCLE

Request Flow

Span Creation

Context Propagation

Parent & Child Relationships

SAMPLING STRATEGIES

Head Sampling

Tail Sampling

Intelligent Sampling

OPENTELEMETRY INTEGRATION

OTel Architecture

OTel Collector

Datadog Exporters

Native vs OTel Instrumentation

03 • Service Dependency Analysis

APM & ADVANCED KPIS WITH DATADOG
SERVICE MAPS

Application Performance Engineering

Investigate latency and errors at depth, quantify performance through core KPIs, and profile code down to the hotspot.

3 Modules · 3 Hands-On Sessions

04 • Advanced Trace Analytics

TRACE EXPLORER

Trace Search & Analytics

Filtering

Aggregation

Grouping

LATENCY ANALYTICS

P50

P75

P90

P95

P99

Slow Transaction Analysis

ERROR ANALYSIS

Error Tracking

Exception Analysis

Failure Correlation

HANDS-ON LAB

→ Trace Investigation

→ Slow Request Analysis

05 • Advanced Performance KPIs

AVAILABILITY KPIS

Uptime %

Availability %

Error Rate

PERFORMANCE KPIS

Response Time

Throughput

Requests / Second

Transactions / Second

RELIABILITY KPIS

MTTR

MTTD

MTBF

USER EXPERIENCE KPIS

Apdex Score

User Satisfaction Index

Session Success Rate

HANDS-ON LAB

APM & ADVANCED KPIS WITH DATADOG

→ Create KPI Dashboards

Business KPI Monitoring & Digital Experience

Connect telemetry to business outcomes, then monitor the real and simulated user journey end to end.

3 Modules · 3 Hands-On Sessions

07 • Business Observability

FOUNDATIONS

Technical vs Business Metrics

Business KPI Monitoring

KPI Instrumentation Strategy

E-COMMERCE KPIS

Cart Abandonment Rate

Conversion Rate

Checkout Latency

Revenue per Minute

BANKING KPIS

Transaction Success Rate

Payment Processing Time

Failed Transaction %

SAAS KPIS

Active Users

Feature Adoption

Subscription Conversion

HANDS-ON LAB

→ Create Business Metrics

→ Build Business Dashboards

08 • Real User Monitoring (RUM)

EXPERIENCE MONITORING

User Journey Monitoring

Frontend Performance

Browser Analytics

CORE WEB VITALS

LCP

FCP

CLS

INP

SESSION INSIGHT

Session Replay

User Segmentation

Reliability, AIOps & Executive Reporting

Design reliability as a discipline — SLIs, SLOs and error budgets — bring AIOps into the loop, and prove mastery through a full-stack capstone.

3 Modules · 3 Hands-On Sessions

10 • SLO Engineering & Reliability

SRE FOUNDATIONS

SRE Principles

Error Budgets

Reliability Framework Design

RELIABILITY CONCEPTS

SLI

SLO

SLA

SLO DESIGN PATTERNS

Availability SLO

Latency SLO

Throughput SLO

BUDGET & BURN

Error Budget Tracking

Burn Rate Monitoring

HANDS-ON LAB

- Create Availability SLO
- Create Latency SLO
- Error Budget Dashboard

11 • Intelligent Alerting & AIOps

ALERT ENGINEERING

Static Thresholds

Dynamic Thresholds

Anomaly Detection

Forecasting

Intelligent Alerting

INCIDENT ANALYTICS

Alert Noise Reduction

Incident Correlation

Root Cause Analysis

AIOps & WATCHDOG

Introduction to AIOps

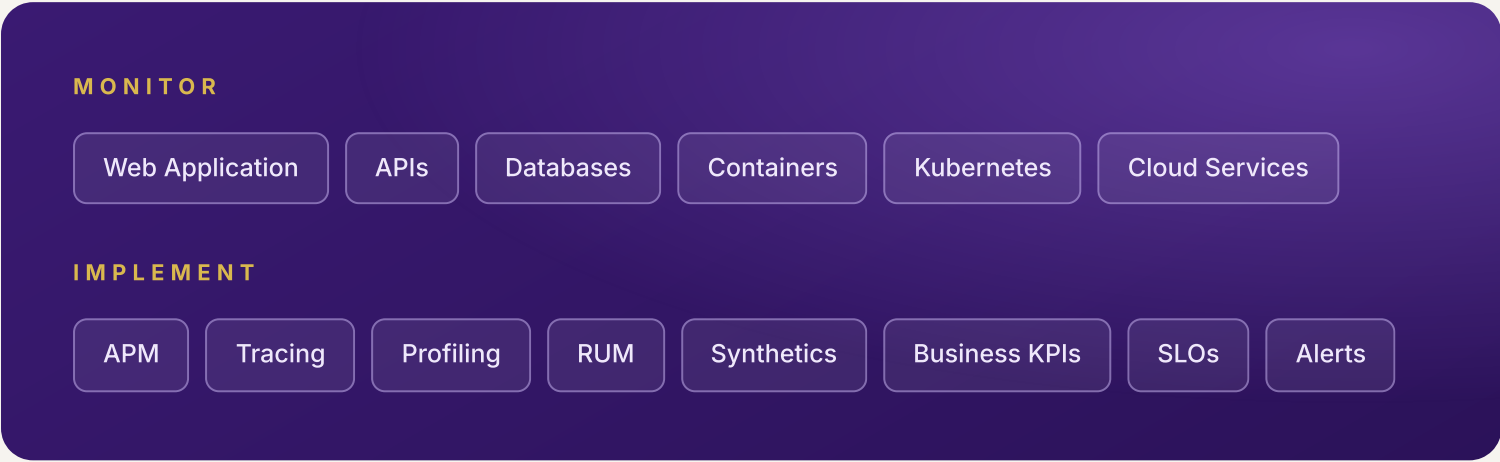
Datadog AI Features

Watchdog

Root Cause Identification

End-to-End Application Monitoring

Bring every discipline together on one production-style system — from web tier to Kubernetes — and prove full observability across signals, reliability, and business outcomes.



CAPSTONE DELIVERABLES

- A

Instrumented full-stack system

Traces, profiles, RUM and synthetics wired across every tier of a live application.
- B

Reliability model

SLIs, SLOs and error budgets defined, with burn-rate alerting that cuts the noise.
- C

Executive narrative

A single reporting suite that translates every signal into business-ready insight.

Advanced KPI Catalogue

A consolidated reference of the metrics engineered across the program — spanning application performance, reliability, user experience, business value, operations, and AI-driven signals.

APPLICATION KPIS

01

- Request Rate
- Throughput
- Latency
- Error Rate
- Availability
- Apdex

RELIABILITY KPIS

02

- MTTR
- MTTD
- MTBF
- Error Budget Consumption
- Burn Rate

USER EXPERIENCE KPIS

03

- LCP
- FCP
- CLS
- INP
- Session Duration
- Session Success Rate

BUSINESS KPIS

04

- Revenue per Minute
- Transaction Success Rate
- Checkout Success Rate
- Conversion Rate
- Active Users
- Customer Satisfaction

OPERATIONS KPIS

05

- Incident Volume
- Alert Accuracy
- Deployment Success Rate
- Change Failure Rate
- Mean Recovery Time

AI-DRIVEN KPIS

06

- Anomaly Frequency
- Root Cause Accuracy
- Predictive Failure Score
- Service Health Score

How the 4-day path is engineered

The four-day program preserves the full curriculum of the extended track. Days 1–3 run unchanged; the reliability and AI-driven days are consolidated into a single culminating day, merging closely related modules without losing depth.

- 1

Days 1–3 — unchanged

Foundations & tracing, performance engineering, and business/digital experience keep their full module set.
- 2

SRE + Advanced SLO » SLO Engineering & Reliability

Reliability principles and SLO design patterns combine into one comprehensive module.
- 3

Alerting + AI Observability » Intelligent Alerting & AIOps

Alert engineering, incident analytics, and Watchdog/AIOps taught as one continuous flow.
- 4

Executive Dashboards + Capstone » Executive Reporting & Capstone

Reporting frameworks feed directly into the full-stack capstone that closes the program.

15→12

MODULES

5→4

DAYS

30+

LABS RETAINED

100%

KPI COVERAGE

OUTCOMES

Performance, Measured by Outcomes

By the close of this program, participants can instrument, trace, and profile applications end to end — and translate that telemetry into reliability SLOs, business KPIs, and executive-ready insight.

APM & ADVANCED KPIS WITH DATADOG · STACKMASTER TECH

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